UNIT-VI

Electromagnetism

PART-A

1.Define Poynting vector in an electromagnetic field. **RTU2022**

2. Define Divergence of electrostatic field and its physical significance. **RTU 18,19,22**

3. State faradays law and Biot- Savart Law. **RTU2022 .23**

4. What do you mean by Maxwell's equation. **RTU 2019**

5.Explain Biot-Savart Law. **RTU 2019,21,22**

PART-B

1. State and explain Lorentz force and Faraday's law of electromagnetic induction. **RTU 2022**

2. Divergence and curl of static magnetic field. **RTU 2021**

3. Derive Poisson's equation and Laplace equation **RTU 2019**

4. Write short notes on :

(a) Displacement current **RTU 2018,19**

(b) Poynting Vector **RTU 2018,19,21,22**

5. State and explain Lorentz force and Faraday's law of electromagnetic induction. **RTU 2022**

6.If $E = 20\hat{i} + 10x^2\hat{j} + 4y\hat{k}$ then find curl at (2,4,6).

7.(a)Calculate the amplitude of displacement current density in air near a car antenna where the

$E=60\cos(6.277\times10^8 t-2.092)\hat{a}_z$ V/m **RTU 2018**

(b)For the material conductivity is 10^{-3} mho/m and relative permittivity $\varepsilon_r = 3$, determine conduction current density if $E= 8\times10^{-6}\sin(6\times10^9 t)$ V/m

9.If $A= 20x^2\hat{i}+10y^2\hat{j}+4\hat{k}$ then find divergence at point (0,1,0).

PART-C

1.(a) Write the Maxwell's equation in their differential form and use them to deduce the integral form of the equation.Briefly explain the physical importance of the maxwell's equation. **RTU 2022**

(b) What is displacement current Jd ? Explain how Maxwell used the continuity equation to introduce the term Jd in order to modify Ampere's law,also how did the introduction of the term Jd revolutionize physics. **RTU 2022**

2(a) Define poynting vector and derive Poynting theorem. **RTU 2022,23**

(b) State Ampere's Circuital law and using Maxwell's correction,derive fourth Maxwell's equation.**RTU 2022**

3(a) Derive Maxwell's equation in differential form. **RTU 2021**

(b) write a short note on a Poynting Vector.

4. What are Maxwell's equations? Derive Maxwell's equation in an isotropic medium and in free space. **RTU2019**

5.Explain and derive Maxwell's equations for the time varying electric fields in differential and integral form. Also give the physical significance. **RTU 2018**